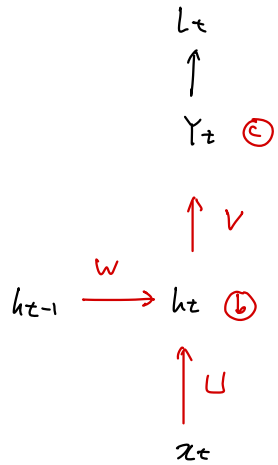
Vanilla RNN

$$\begin{cases} h_t = \sigma(W h_{t-1} + U x_t + b) \\ Y_t = V h_t + c \end{cases}$$

$$\begin{cases} L_t = \frac{1}{2}(Y_t - z_t)^2 \\ L = \sum L_t \end{cases}$$



$$\frac{\partial L}{\partial V} = \sum_t \frac{\partial L_t}{\partial Y_t} \cdot \frac{\partial Y_t}{\partial V}$$

$\frac{\partial L_t}{\partial Y_t} = Y_t - z_t$, $\quad \frac{\partial Y_t}{\partial V} = h_t$

$$\frac{\partial L}{\partial c} = \sum_t \frac{\partial L_t}{\partial Y_t} \cdot \frac{\partial Y_t}{\partial c}$$

$\frac{\partial L_t}{\partial Y_t} = Y_t - z_t$, $\quad \frac{\partial Y_t}{\partial c} = 1$

Backprop Through Time

$$\frac{\partial L}{\partial W} = \left(\frac{\partial L_1}{\partial h_1} \cdot \frac{\partial h_1}{\partial W}\right) + \boxed{\left(\frac{\partial L_2}{\partial h_2} \cdot \frac{\partial h_2}{\partial W}\right) + \left(\frac{\partial L_2}{\partial h_2} \cdot \frac{\partial h_2}{\partial h_1} \cdot \frac{\partial h_1}{\partial W}\right)}$$

$$\frac{\partial h_L}{\partial W} = \sum_{\ell=1}^{L} \left(\frac{\partial h_{\ell+1}}{\partial h_\ell} \cdots \frac{\partial h_2}{\partial h_{L-1}}\right) \cdot \frac{\partial h_\ell}{\partial W}$$

$\frac{\partial h_2}{\partial h_{L-1}} = (1 - h_L^2) \cdot W$, $\quad \frac{\partial h_\ell}{\partial W} = (1 - h_\ell^2)\, h_{\ell-1}$

Note

$(\tanh(x))' = 1 - \tanh^2(x)$

$\left|\frac{\partial h_s}{\partial h_{s-1}}\right| > 1$ then $\quad 100 \to \infty \quad \Rightarrow$ clipping : $(\text{threshold}) \cdot \frac{g}{\|g\|}$

$\left|\frac{\partial h_s}{\partial h_{s-1}}\right| < 1$ then $\quad 100 \to 0$

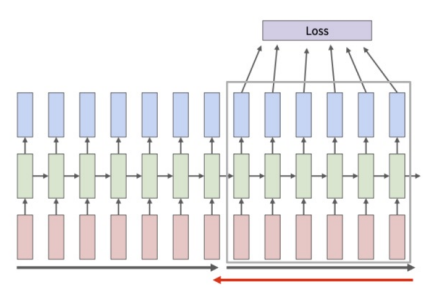
$$\frac{\partial L}{\partial h_\ell} = \frac{\partial L}{\partial h_{\ell+1}} \cdot \frac{\partial h_{\ell+1}}{\partial h_\ell}$$

$\approx I$ then

grad can flow $\begin{cases} \text{LSTM} \\ \text{GRU} \end{cases}$

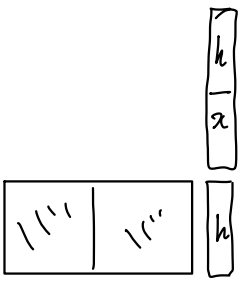
Note.

Truncated Backprop.



Vanilla RSTM

$$h_t = \tanh\left(W\begin{pmatrix} h_{t-1} \\ x \end{pmatrix}\right)$$

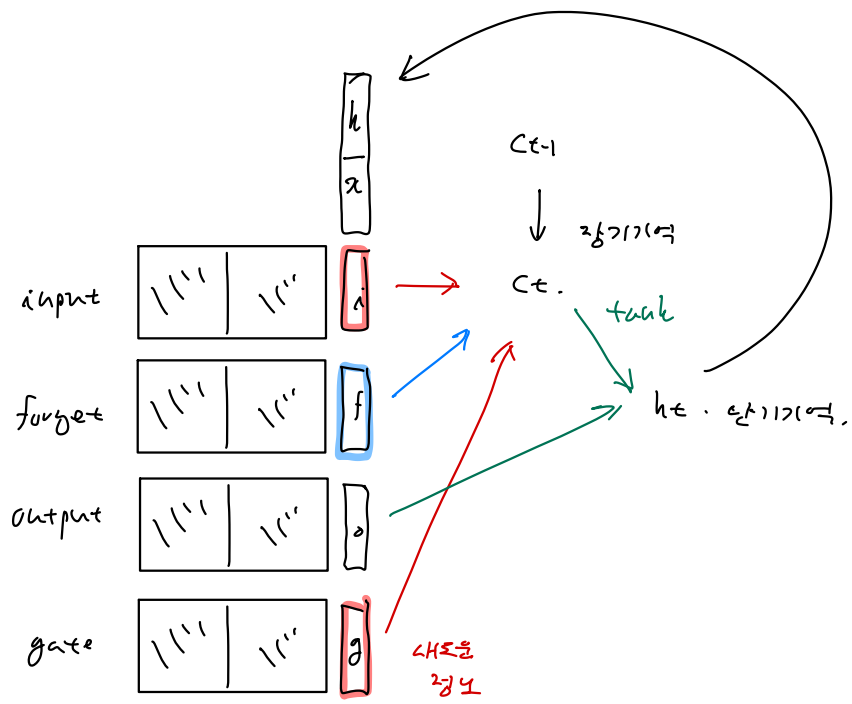


LSTM

$$\begin{pmatrix} i \\ f \\ o \\ g \end{pmatrix} = \begin{pmatrix} \sigma \\ \sigma \\ \sigma \\ \tanh \end{pmatrix} W \begin{pmatrix} h_{t-1} \\ x_t \end{pmatrix}$$

$$c_t = f \odot c_{t-1} + i \odot g$$

$$h_t = o \odot \tanh(c_t)$$



- $\dfrac{\partial L}{\partial c_t} = \dfrac{\partial L}{\partial c_{t+1}} \dfrac{\partial c_{t+1}}{\partial c_t}$

- $c_{t+1} = f_{t+1} \odot c_t + i_{t+1} \odot g_{t+1}$

  - $\dfrac{\partial c_{t+1}}{\partial c_t} = f_{t+1} + (\cdots)$

  - 중요한 정보에 대해 $f_{t+1} \approx 1$

uninterrupted grad flow

$h_t = LSTM(z_t, h_{t-1}, c_{t-1}; \theta)$

Note. One-Hot Encoding

"a" [0 0 1]

"man" [0 1 0]

∴

- $z_t = W_{emb} x_t$

  one-hot

- $\hat{x}_{t+1} = W_d h_t + b_d$

  "다음 단어로 등장할 확률"

---

training $\sum_{t=1}^{T} \mathcal{L}(\hat{x}_t, x_t)$

cross-entropy loss $-\sum p(x) \log q(x)$

inference $\begin{bmatrix} \text{Greedy} \\ \text{argmax } S(x_t) \end{bmatrix}$ or $\begin{bmatrix} \text{stochastic} \\ x_t \sim S(x_t) \end{bmatrix}$

... <EOS> ⇒ 생성 종료.

---

img to text

CNN

$f_\theta(y \mid f_\phi(x))$

RNN

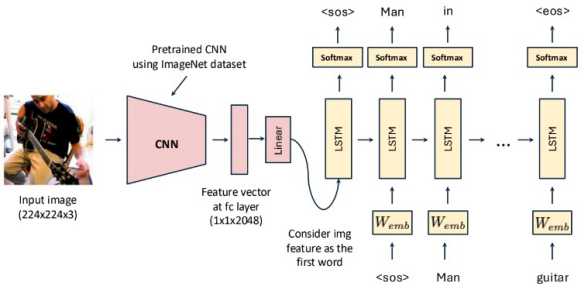
"use feature vector as first word"

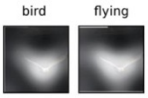
"ho 정보를 잃어버린다면?"

개선 ① 매 순간 단어에게나 함께 입력

개선 ② Attention

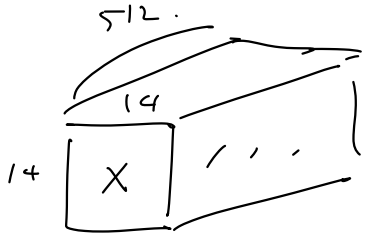
: 단어를 생성할 때 관련된 이미지의 특징 부분을 확인!





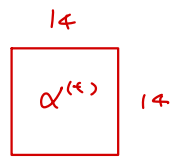
Spatial Attention

- X Conv5 (14x14x512)



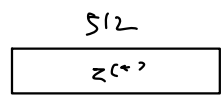
- score $e^{(t)} = f_{att}(X, h_{t-1})$

각 조각은 "a bird"와 얼마나 관련 있는가?



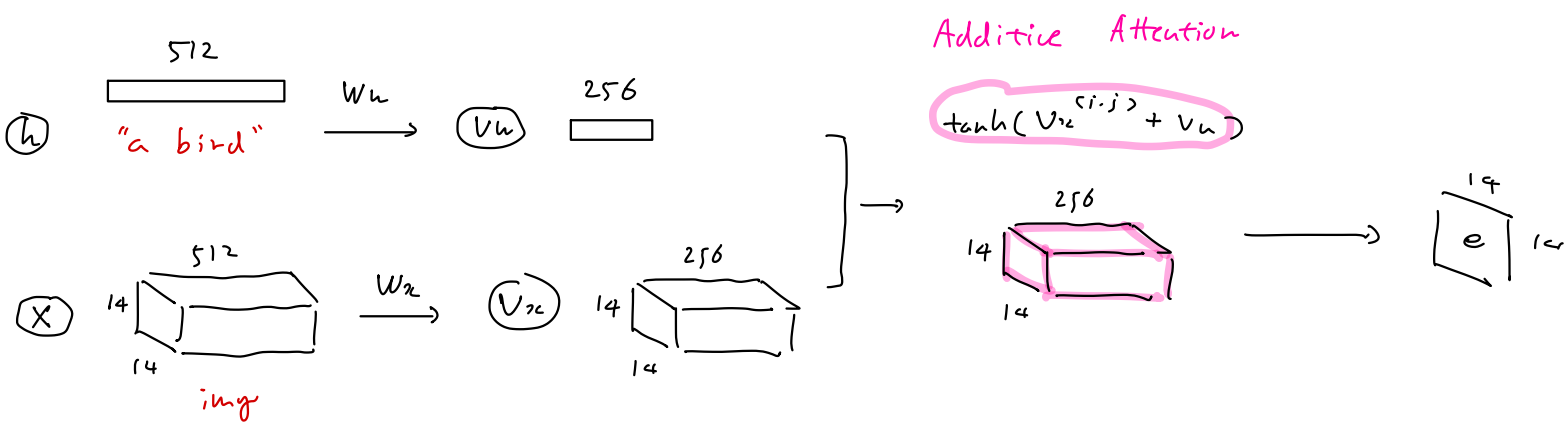
Normalize $\alpha^{(t)} = \text{softmax}(e^{(t)})$

Context $z^{(t)} = \sum_{i,j} \alpha^{(t)}_{i,j} x_{i,j}$

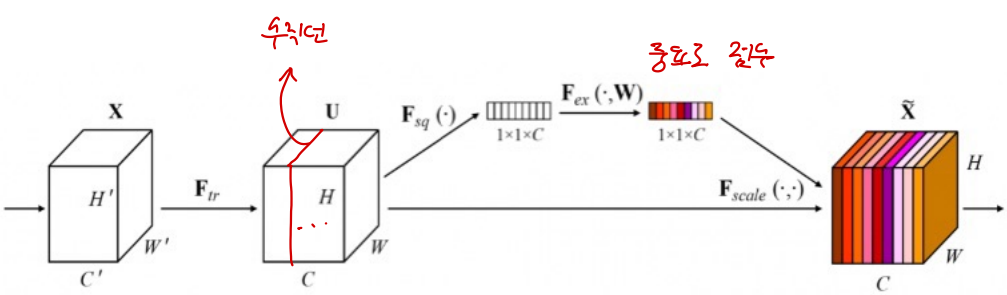


- $h^{(t)} = \text{LSTM}(h_{t-1}, W_{emb}\, y_t, z^{(t)})$

Simple Spacial Attention



Squeeze-and-Excitation Net : Channel Attention.

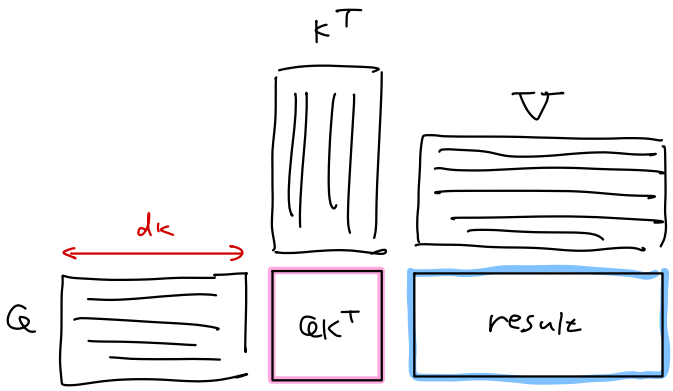


$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right) \cdot V$$

softmax $\frac{e^{z_j}}{\sum e^{z_j}}$

train ↓

- Query $= XW_q$: $n \times d_k$
- Key $= XW_k$: $n \times d_k$
- Value $= XW_v$: $n \times d_v$



Self-Attention vs Cross-Attention

$Q, K, V \Rightarrow X$

$Q \rightarrow Y$ (decoder)

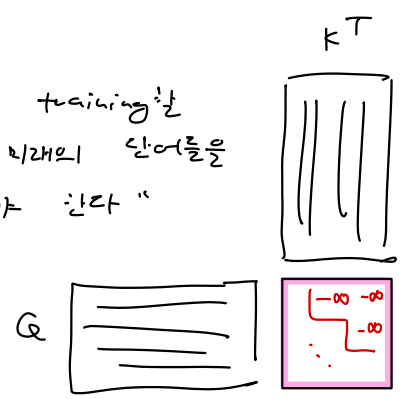
$K, V \Rightarrow X$ (encoder)

예시

기계번역	이미지 캡션
Q : 번역할	Q : h_{t-1}
K, V : 원문	K, V : 이미지 특징

Masked Self-Attention

"decoder를 training 할 때에는 미래의 단어들을 가려야 한다"



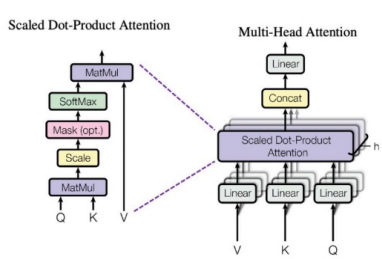
i^{th} row < j^{th} column then $-\infty$

positional encoding

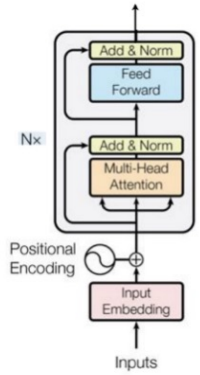
: $x_j + p_j$ or concat (x_j, p_j)

$$2d \begin{bmatrix} \sin(w_1 \times pos) \\ \cos(w_1 \times pos) \\ \ddots \\ \sin(w_d \times pos) \\ \cos(w_d \times pos) \end{bmatrix} \quad \text{where} \quad w_k = \frac{1}{10000^{k/d}}$$

Multi-Head Attention : h개의 다양한 W_q, W_k, W_v를 이용



Transformer Encoder.



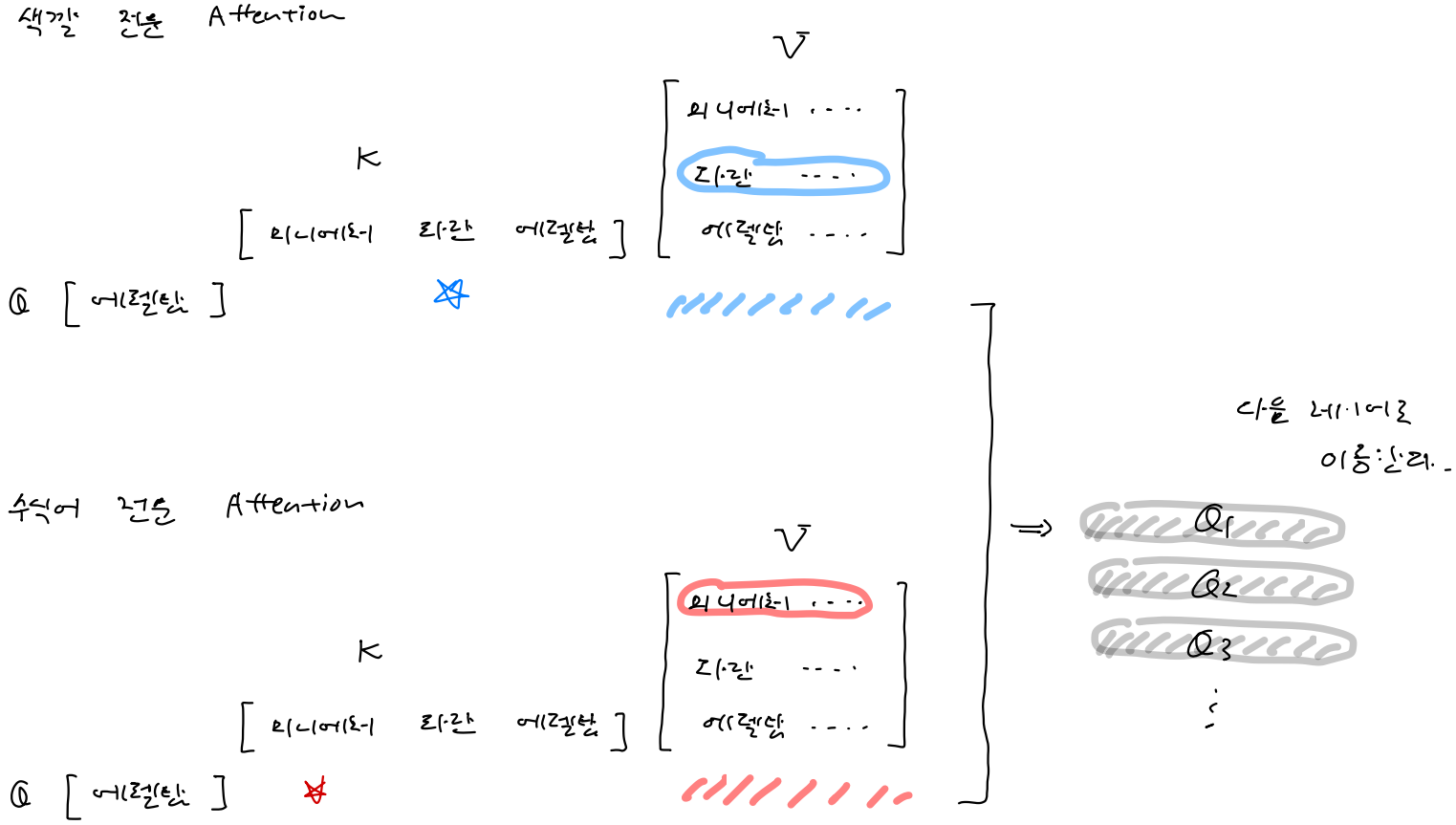
→ LayerNorm $(h + FF(h))$

→ LayerNorm $(h + Att(h, h, h))$

→ $FF(x_i)$

"Attention is All You Need"

Attention & Transformer 의 이해



Linearization $\quad \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)$ is too big!

$\text{sim}(q, k)$

$$= \exp\left(\frac{q^T k}{\sqrt{D}}\right) \longrightarrow \text{ker}(q, k) = \phi(q)^T \phi(k)$$

리장 순간

$$\text{then} \quad \tilde{V}_i = \frac{\sum_{j=1}^{N} \text{ker}(q_i, k_j) \cdot v_j}{\sum_{j=1}^{N} \text{ker}(q_i, k_j)} = \frac{\phi(q_i)^T \sum_{j=1}^{N} \phi(k_j) \cdot v_j}{\phi(q_i)^T \sum_{j=1}^{N} \phi(k_j)}$$

미리 계산 $(r \times d_v)$

리장 순간.

미리 계산. (d_v)

Normalization.

배치 채널 호이다니

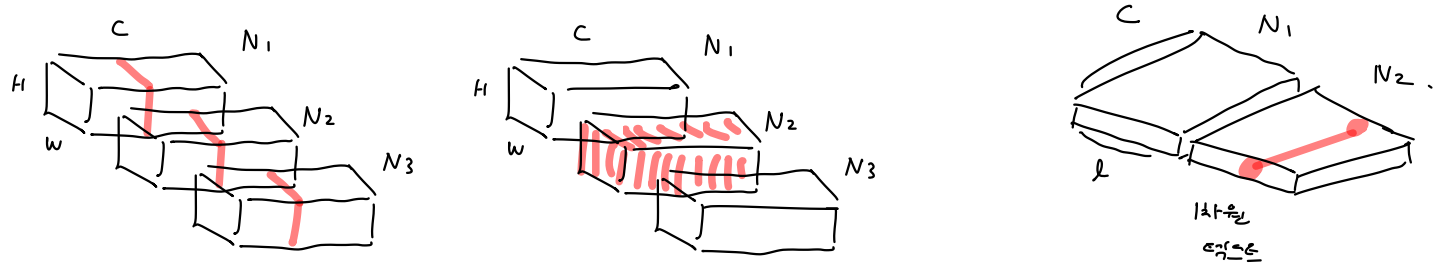
$$X \in \mathbb{R}^{N \times C \times (H \times W)}$$

or L

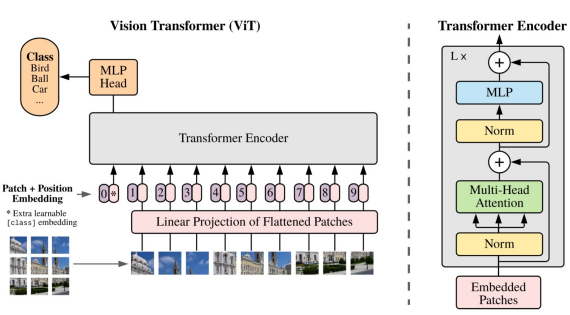
$$LN(x_i) = \beta \cdot \frac{x_i - \mu}{\sqrt{\sigma^2 + \varepsilon}} \cdot \gamma$$

"ℓ을 가변적일 수 있다!"

① Batch Norm (CNN) ② Layer Norm (CNN) ③ Layer Norm (Transformer)



$$\mu_c = \frac{1}{NHW} \sum_{n,h,w} x_{n,c,h,w} \qquad \mu_n = \frac{1}{CHW} \cdot \sum_{c,h,w} x_{n,c,h,w} \qquad \mu_{n,l} = \frac{1}{C} \cdot \sum_{c} x_{n,l,c}$$



- 이미지를 N개로 잘랐다. (patching)

- $z_0 = [\overbrace{x_{class}}^{D\text{차원}} ; \overbrace{x_p^1 E}^{D\text{차원}} ; \cdots ; \overbrace{x_p^N E}^{D\text{차원}}] + E_{pos}$

- $$\begin{cases} z'_\ell = MSA(LN(z_{\ell-1})) + z_{\ell-1} \\ z_\ell = MLP(LN(z'_\ell)) + z'_\ell \end{cases}$$

- 맨 앞에 있는 D차원 정보만 가져옴.

GELU $= x \cdot \Phi(x)$

가우시안 분포의

누적확률분포함수

CNN

: 작은 filter가

sliding window로

feature map을 만든다.

✓ 계층적 구조

: 엣지, 선 → 눈, 코, 나무

✓ 지역적 처리

: 필터 크기에 의존

ViT

: 이미지를 patch로 나눈 뒤

{ flattening
position encoding } 을 거쳐

transformer에 넣는다.

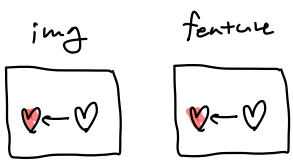
✓ 전역적 처리

: 대체 안의 관계를

한번에 계산

강한 사전가정

- locality
- Translation Equivariance

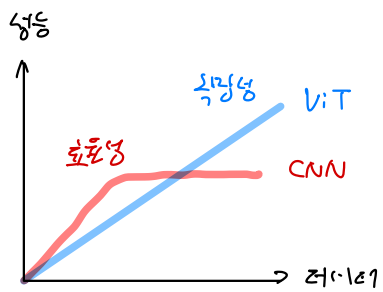


$F(T(x)) = T(F(x))$

약한 사전 가정

- 無 locality
- 無 Translation Equivariance

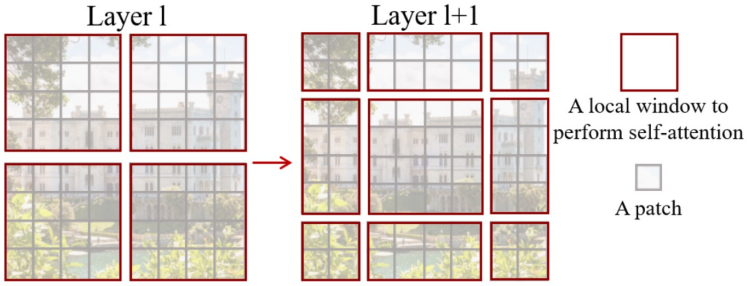
⟸ position encoding



Note.

translation invariance

$F(T(x)) = F(x)$



① ViA vs W-MSA.

Window 안에서만 Attention.

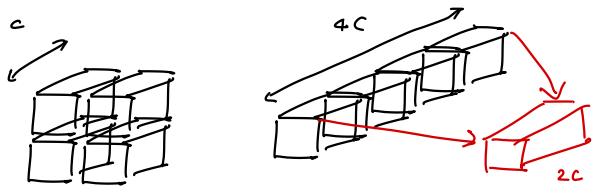
$$\begin{cases} \Omega(MSA) = 4hwC^2 + 2(hw)^2 \cdot C \\ \Omega(W\text{-}MSA) = \quad \cdot\cdot \quad + 2M^2 hwC \end{cases}$$

② Shifted window

: 윈도우의 경계를

M/2 만큼 줄임.

③ Patch Merging



like pooling

in CNN.